



Senior Data Scientist — Take-Home Case Study

Role: Senior Data Scientist

Company: Office Timeline

Format: Take-home (3–4 hours core + 60-minute live presentation and Q&A)

Distribution: Provided one week before the interview

Scenario

Office Timeline sells a Microsoft PowerPoint add-in on a B2B SaaS model. Users download and install it before evaluating whether to purchase. PLG customers discover the product through search (PPC and organic), download a free version or start a 14-day trial, install it, and — if it meets their needs — convert to a paid subscription.

The marketing team tracks this motion through a free-to-paid (F2P) funnel dashboard.

Since we started using the install tracker (August 25, 2025)

Monthly Funnel Cohort Detail

Conversion Rate = Paid / Complete

Complete = Completed the installation after the contact creation date in HubSpot

Paid = Completed a transaction after the contact creation date in HubSpot

Contact Created in HubSpot → **Start Install** → **Complete Install** → **Open Add-in** → **Purchased**

Cohort	Month	Contact Created	Start	Complete	Open	Paid	Contact : Start	Start : Complete	Complete : Paid	Avg. Days (Contact : Paid)	Contact : Paid
2026-04-01	April 2026	8,440	4,707	3,745	3,165	237	55.8%	79.6%	6.3%	0.92	2.8%
2026-03-01	March 2026	17,311	9,064	5,668	5,127	528	52.4%	62.5%	9.3%	2.24	3.1%
2026-02-01	February 2026	14,444	7,649	4,565	4,356	607	53.0%	59.7%	13.3%	4.28	4.2%
2026-01-01	January 2026	11,721	7,297	4,456	4,276	618	62.3%	61.1%	13.9%	6.09	5.3%
2025-12-01	December 2025	8,954	5,944	3,774	3,542	460	66.4%	63.5%	12.2%	10.77	5.1%
2025-11-01	November 2025	12,586	7,570	5,009	4,776	608	60.1%	66.2%	12.1%	12.70	4.8%
2025-10-01	October 2025	14,580	9,327	6,099	5,793	696	64.0%	65.4%	11.4%	9.71	4.8%
2025-09-01	September 2025	14,364	9,089	5,809	5,402	654	63.3%	63.9%	11.3%	12.35	4.6%
2025-08-01	August 2025	3,133	2,000	1,307	1,227	141	63.8%	65.4%	10.8%	13.64	4.5%
Total		105,532	62,647	40,432	37,664	4,549	59.4%	64.5%	11.3%	8.19	4.3%

The dashboard reports two conversion rates: **4.3% Contact:Paid** and **11.3% Complete:Paid** (Aug 2025–Apr 2026, reseller transactions excluded, all-time conversion window). The dashboard prominently highlights the 11.3% Complete:Paid number as the conversion rate (formula shown: `Conversion Rate = Paid / Complete`).

The CFO believes this number is not the right number to be reporting or optimizing against. She hasn't told you why. Your job is to find out.

You have been given:

- A CSV export of the F2P funnel view (~104,000 rows, Aug 2025–Apr 2026). Emails have been anonymized; corporate domains are also anonymized (e.g., `user-7a3f@company-abc12.com`); personal email domains (gmail, yahoo, etc.) are preserved as-is.
- The SQL definition of the view the CSV was generated from.
- Schemas for the key silver-layer tables the view draws on: `pes_subscriptions` , `pes_billingdata` , `hs_otl_contacts` .
- Gold-layer NULL-rate stats for the current `dim_pes_*` tables, illustrating the identity resolution coverage problem you'll address in Part 3.
- A data dictionary — read this before starting.

A note on February 2026: Marketing ran an A/B test requiring email validation before download. The test was not properly instrumented and is known to have affected funnel data for that month in ways that are difficult to fully account for.

Three numbers to ground your analysis:

- Current dashboard Contact:Paid: **4.3%** (reseller transactions excluded, all-time conversion window)
- Current dashboard Complete:Paid: **11.3%** (the headline conversion rate as reported on the dashboard)
- Assume ARPU of **\$200** for any unit economics calculations

Part 1 — Analytical Take-Home

This is the core of the exercise. A strong candidate can complete it in 3–4 hours.

1a. What is the right conversion rate?

These rates are a starting point, not an answer. Working from the CSV, the data dictionary, and the view SQL:

1. **Audit the denominator.** Who is in the contact population? Identify any contact types or source categories that should not count as genuine prospective new customers — contacts who may appear in the funnel for reasons rather than because they independently discovered and chose to evaluate the product. For each one you identify: how many contacts are affected, what share of paid conversion does it represent, and what does C → Paid look like with and without it?
2. **Audit the conversion window.** The dashboard uses an all-time conversion window: a contact is “paid” if they ever converted, regardless of how long it took. Propose and justify an alternative window. Show how the rate changes.
3. **State the number you’d report.** Give one clean C → Paid rate with a precise definition of denominator and conversion window. Explain in one paragraph why your rate is more defensible than the dashboard rates.

1b. Trend and channel

Using the contact population and conversion window from 1a:

- How has C → Paid trended month-over-month over the period in the data?
- Does the trend look different by traffic source or acquisition channel?
- Is there anything in the data that would change how you'd allocate the marketing budget? Use the spend figures below to size your answer.

Month	PPC Spend	Total Marketing Spend
Sep 2025	\$145,080	\$265,844
Oct 2025	\$150,839	\$259,284
Nov 2025	\$131,709	\$242,105
Dec 2025	\$100,545	\$225,893
Jan 2026	\$158,916	\$285,625
Mar 2026	\$187,971	\$325,782

February 2026 is not included. April 2026 data was extracted in mid-April — the paid conversion window is still open for that month; treat April as preliminary.

Part 2 — Live Presentation

25–30 minutes presentation, 30 minutes Q&A. We will interrupt during both.

Section A — Funnel findings (15–20 min)

Present your findings from Part 1 to a mixed room: the CFO, the VP Marketing, and the VP Customer Success. Assume they have not read your write-up.

You should be able to:

- State the right conversion rate(s) and explain why they differ from the dashboard's 4.3% and 11.3%, in plain language
- Show the trend and channel story, with key findings called out
- Make a clear recommendation on what to fix and what to leave alone — we will push back on your prioritization

Section B — Business partnership (10 min)

You will work closely with Product, Marketing, Sales, Customer Success, and Finance. For **two functions of your choice**:

- Name the two or three data questions you believe that leader most needs answered right now
- For each: what data feeds it, what cadence does it need, and is it operational or analytical?
- Which of your chosen questions is most underserved by the current data platform, and why?

We will ask about the remaining functions in Q&A.

What distinguishes a strong presentation:

- You lead with the business question, not the methodology
- You are precise about what you know vs. what you're inferring
- You don't overstate confidence where the data is genuinely ambiguous

Part 3 — Platform Design

Tests whether you can design infrastructure that reduces your own future workload and enables the business to be self-sufficient.

Context

The CSV was exported from `reporting.view_funnel_dashboard`, a reporting view built directly on silver-layer tables. It does its own identity resolution, applies its own filters, and computes its own funnel flags — bypassing the gold layer entirely. This means the clean C → Paid metric you computed in Part 1 doesn't exist in a governed form anywhere in the data model. The next analyst may compute it differently. The CFO can't pull it herself.

A related problem: customer, subscription, and seat-assignment data lives in PES, which CDCs into the lakehouse. HubSpot holds marketing and CRM contacts. These identity spaces are not reliably unified in the current gold layer — a spot-check on a large enterprise customer found 849 seats assigned in PES silver but only 53 resolvable in gold, because the silver → gold transform fails to join PES user GUIDs to application-layer identities for most users.

Both problems have the same root cause: source-system unification is happening in the reporting layer, where it doesn't belong.

The design task

Propose the gold-layer additions that address both problems — metric materialization and user identity resolution. Scope your answer to what's necessary to make the funnel metric trustworthy and the user identity resolvable; this is not a full gold layer rewrite.

Cover:

- What belongs in gold vs. computed at query time, and why
- Sketched table design (column names, grain, derivation logic — no full DDL needed)
- How the clean C → Paid definition is preserved so the next analyst gets the same number, not a drift variant

- A `dim_user` design that unifies PES GUIDs, HubSpot contact IDs, email, and email domain — including your resolution strategy and how unresolvable users surface to the consumer rather than silently disappear
- What the self-serve access pattern looks like in practice — not just for this metric, but as a pattern. Who can query what, how do they find it, and what prevents divergent answers from proliferating across tools and teams?

Ground your proposal with one concrete artifact — your choice of a SQL prototype, table definition, metric spec with governance wrapper, or lineage diagram.

Part 4 — Above and Beyond (*optional*)

Only attempt this if you've completed Parts 1–3 to a standard you're satisfied with.

The gold layer gap in Part 3 is one instance of a broader pattern: reporting views that bypass gold, metrics computed differently in different places, business users filing tickets for numbers that should be self-serve.

In 400–600 words:

- What organizational pattern produces this outcome?
 - What does a data team of two (one data engineer, one data scientist) do first to break the cycle?
 - What would you explicitly NOT do in the first 90 days, and why?
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